

Denish Goyal

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SUMMARY

Software Engineer with strong **computer science fundamentals** and hands-on full-stack development experience across the SDLC. Proficient in **Python**, React.js, and Node.js with solid grounding in **object-oriented design**, **TCP/IP networking concepts**, and **cloud-scalable** architectures (AWS, Docker, Kubernetes, Terraform). Passionate about **UI/UX design**, comprehensive **testing**, and leveraging AI coding agents to accelerate development and deliver well-tested, production-ready software through **Agile** delivery.

EDUCATION

University of Petroleum and Energy Studies
B.Tech in Computer Science – **CGPA: 7.73/10**

Aug 2023 – Jul 2027

TECHNICAL SKILLS

Languages:	Python, Go (Golang), C++, JavaScript, C
CS Fundamentals:	OOP, Data Structures, TCP/IP Networking, Routing & Switching Concepts
Databases:	PostgreSQL, MySQL
Frameworks & ORMs:	FastAPI, Node.js, React.js, Prisma ORM, REST APIs
DevOps & Cloud:	AWS (EC2, S3, ECS, EKS), Docker, Kubernetes, Terraform (IaC)
Tools & Testing:	Jenkins, GitHub Actions, Postman, Linux, Git, Debugging Tools

EXPERIENCE

Chirpn IT Solutions *Software Engineering Intern (Remote)* Jun 2025 – Jul 2025

- Strengthened application-layer defenses across **10+ React.js components** by conducting full-stack security reviews spanning frontend **UI** and backend layers—identifying XSS vulnerabilities via SAST scanners, patching input-sanitization logic, and validating fixes through automated **regression tests** within Agile sprint cycles.
- Reduced manual engineering effort by **70%** by building an AI-augmented DevSecOps pipeline using **Python** scripts and the Claude Agent SDK to automate vulnerability detection, severity classification, and remediation tracking across the SDLC.
- Achieved **87% accuracy** in automated triage decisions by designing **evaluation harnesses** over 200+ labeled security alerts, implementing quality guardrails and escalation logic that ensured reliable, **well-tested** AI-agent output throughout the development lifecycle.
- Cut deployment latency by **35%** by automating end-to-end **cloud provisioning** and API rollouts through Terraform (IaC) and GitHub Actions CI/CD pipelines, establishing repeatable infrastructure scaled across AWS (EC2, ECS).

PROJECTS

Custom API Rate Limiter Feb 2026

Go (Golang), TCP/IP, System Architecture

- Achieved **sub-millisecond** decision latency by engineering a concurrent rate-limiting service in Go using the Token Bucket algorithm to throttle abusive traffic and mitigate application-layer DDoS vectors at the **TCP/IP** transport layer.
- Enabled zero-downtime API protection by designing a stateless, **horizontally scalable** architecture with comprehensive **unit and integration tests**, suitable for cloud deployment in automated security-enforcement pipelines.

K8N – Visual IDE for Kubernetes Mar 2026

React.js, ReactFlow, Go (Golang), Kubernetes, Docker, REST APIs

- Engineered a **graph-based visual IDE** for Kubernetes using **ReactFlow** to let users drag-and-drop pods, services, and deployments onto an interactive canvas—translating node/edge graphs into valid K8s manifests via a Go backend with **OOP**-structured service and repository layers.
- Implemented real-time cluster state sync through the Go backend's Kubernetes client-go SDK, enabling users to **visually debug** resource relationships, inspect pod logs, and deploy configurations directly from the IDE with zero manual YAML editing.

ACHIEVEMENTS & ADDITIONAL SKILLS

- **Testing & Debugging:** Strong testing discipline—authored unit, integration, and regression test suites; proficient with debugging tools (GDB, Chrome DevTools, VS Code debugger) and multiple development environments.
- **Open Source & Prototyping:** Contributed to open-source projects and collaborated in Agile hackathon environments to architect and deploy functional prototypes with rapid API integrations.
- **Algorithmic Optimization:** Built a strong foundation in complex data structures and algorithmic efficiency using C++, directly translating to writing highly optimized, scalable application code.